College of Computer Studies and Engineering
Computer Engineering / Electronics Engineering Department

THESIS PROPOSAL / DESIGN PROJECT
REVISIONS COMPLIANCE FORM

PROJECT TITLE: Non-Invasive Cholesterol Screening Device Using Hand Imaging and Artificial Neural Networks		
Program: **Bachelor of Science in Computer Engineering**	**Course/Section: CPE C403 – 401G**	**School Year:** **2025 - 2026**
Subject Adviser: Engr. Rosalina Estacio		
Proponents: **1. Derilo, Ashley Shane B.** **2. Francisco, Florenz Jay B.** **3. Jontilano, John Jelord M.** **4. Villas, Rhynne Charz M.** **Technical Adviser: Richard Ramos, PhD**		

List of Recommendations	Revision/s Made	Page or Part of Project reflected
Revision of Title	*Changed "monitoring" to "screening" and removed "senior citizens.".*	*Title Page*
Specify LED lighting type and specs	*Added detailed LED specifications under Engineering Requirements.*	*Ch. 3.3.3.2 (P.55)*
Clarification in database usage	*The database used is Firebase. It stores the following data:* *User Information, Cholesterol Test Data, and Admin Accounts.*	*Entire Paper*
User App and Admin App	*Added wireframes for both User and Admin applications.*	*Ch. 3.3.3.3 (P.60, 61, 62, 63)*
Alternative preprocessing (besides GLCM or combination)	*Included other possible preprocessing techniques we can use under Engineering Requirements.*	*Ch. 1.2 (P.4)* *Ch. 3.3.3.2 (P.55)*
Consider alternative preprocessing technique	*Included alternative preprocessing techniques under Engineering Requirements.*	*Ch. 1.2 (P.4)* *Ch. 3.3.3.2 (P.55)*
Compare AI models to identify best performer	*Conducted comparison of several AI models and ANN*	*Ch. 1.2 (P.4)* *Ch. 3.3.3.2 (P.55)*

	was determined to be the best performer.	
Clarification in method for testing result validity	*It involves comparing the system's output with actual laboratory test results to verify accuracy, consistency, and reliability, as discussed in the Data Collection and Testing Plan and Procedures sections.*	*Ch. 3.2 (P.41, 42)*
Correct flowchart symbols and flow	*Corrected overall flowchart design, including proper symbols and flow direction.*	*Ch. 3.3.2(P.45,46)* *Ch. 3.3.2.1(P.48, 50)* *Ch. 3.3.2.2(P.51)* *Ch. 3.3.2.3(P.53)*
Add missing image labels	*Added figure labels and corresponding descriptions.*	*Ch. 3.3.3.3(P.57, 58, 59)*
Change TA name on approval sheet to black	*Changed red text color to black.*	*p. II*
Improve app design for result readability	*Presented screenshots individually for clarity.*	*Ch. 3.3.3.3(P.60, 61, 62, 63)*
Specific objective revision	*Revised objectives by removing specific AI model algorithms and preprocessing techniques to allow flexibility in exploring various combinations for achieving the best accuracy.*	*Ch. 1.2 (P.4)*
Adding Hand Texture in RRL	*Added literature and studies related to hand texture (Normal and High).*	*Ch. 2.1.6 (P. 18, 19, 20)* *Ch. 2.2.6 (P. 31, 32, 33, 34, 35)*
Adding Image Preprocessing in RRL	*Added literature and studies related to image preprocessing.*	*Ch. 2.1.7 (P. 20, 21)* *Ch. 2.2.7 (P. 35, 36, 37)*
Missing discussion of the Testing Plans and algorithm performance comparison	*Added a table about Algorithm Comparative Performance Table showing results of various preprocessing methods and AI models with performance metrics and discussion.*	*Ch. 3.4 (P. 67, 68)*
Include a procedure on how to use the device.	*Added Section for the user's guidelines with a clear step-by-step guide on how to use the device.*	*Ch. 3.3.3.4 (P.64, 65)*
List of tables should be in Title Case and Regular Font	*The List of Tables was updated from all caps to*	*p. IV*

	title case with a regular font for improved readability and consistency.	
Consistency of the word "*study*" in the entire paper	*All instances of "Research" were changed to "Study" to maintain consistent terminology throughout the paper.*	*Entire Paper*
Missing software methodology section	*A dedicated Software Methodology section was added under Chapter 3 to describe the approach used in developing the software.*	*Ch. 3.3.4 (P.65)*
Add a discussion about the Software Design Development model	*A discussion on the chosen software development model was included to explain the framework guiding the project.*	*Ch. 3.3.4 (P.65)*
Add Classification Model Discussion	*The Classification Model section was added to describe the algorithm, its structure, and its role in the project.*	*Ch. 3.3.4.1 (P.65)*
Add Software tool/platform discussion	*A discussion on the software tools used was added to justify the selection of technologies for development.*	*Ch. 3.3.4.2 (P.66)*
Add discussion for algorithm and processes	*The algorithm section now explains the preprocessing, model, and steps involved in the system's operations.*	*Ch. 3.3.4.3 (P.68)*
Add discussion on each testing plan and procedures table	*A discussion on the testing plan was included.*	*Ch. 3.4 (P.69, 70, 71, 72, 73, 74)*
Add discussion on AI model performance metrics. Discuss and add necessary formulas with discussion and tables	*Added a discussion of AI model performance metrics.*	*Ch. 3.4 (P.73,74)*
Add comparison of model performance vs previous studies	*A comparison table and discussion were added to test the model's performance against previous studies.*	*Ch. 3.4 (P.73,74)*
Revise comparative table of the testing plans and algorithm performance comparison	*The comparative table was revised for clarity, showing differences in testing strategies and algorithm effectiveness.*	*Ch. 3.4 (P.72,73)*

C Complied NC Not Complied	Examination Committee	Signature	Date
Complied	ENGR. ROSALINA ESTACIO		11-06-25
Complied	ENGR. AMOR ELLARES		11-07-2025
Complied	ENGR. JOSHUA GULMATICO		Nov. 16, 2025

REVIEWED BY:

PROGRAM CHAIR : ENGR. ROSALINA ESTACIO DATE : 11-06-25

DEAN : ENGR. LIZA R. REYES DATE : _____